

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology

University Examination May 2011

EE-202 Electrical Power Generation

Date: 18.05.11, Wednesday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

Section – 1

- Q – 1 (a) List out the major auxiliaries used in thermal power station 06
- (b) Draw the schematic arrangement of thermal power station and give the name of various cycles of modern thermal power plant 05

- Q – 2 (a) How the hydro electric plants are classified? Discuss each plant briefly 06
- (b) Explain different types of turbines used in hydro electric power plants 06

OR

- (b) With a neat schematic diagram, explain the working of gas turbine power plant 06
- Q – 3 (a) Explain the working principle of MHD power plant with neat schematic diagram 06
- (b) Write a short note on nuclear reactors 06

OR

- (b) Discuss the factors to be considered for the site selection of nuclear power plant 06

Section – 2

- Q – 4 (a) What is fuel cell? Describe the performance characteristic of fuel cell 06
- (b) What is solar pond? Explain solar pond electric power plant with neat sketch 05

- Q – 5 (a) Explain the working principle of ocean thermal energy converter of closed cycle type and open cycle type 06
- (b) Describe the working principle of tidal power generation. List out the difficulties in the development of tidal power generation 06

OR

- (b) Draw the schematic arrangement of diesel power station and explain the essential elements of it 06
- Q – 6 (a) List out the different types of cycles used in geothermal power generation. Explain any one cycle for power generation by geothermal energy 06
- (b) Derive the power versus wind velocity characteristic and define power coefficient 06

OR

- (b) Which type of hybrid energy generation can be possible with wind power generation? Explain any one of them. 06